

# **FNCE 5352 — Lecture 5 (Part 1): Classification & Logistic Regression**

# Classification

- Qualitative variables take values in an unordered set  $\mathcal{C}$ , such as:  
 $\text{eye color} \in \{\text{brown}, \text{blue}, \text{green}\}$   
 $\text{email} \in \{\text{spam}, \text{ham}\}.$
- Given a feature vector  $X$  and a qualitative response  $Y$  taking values in the set  $\mathcal{C}$ , the classification task is to build a function  $C(X)$  that takes as input the feature vector  $X$  and predicts its value for  $Y$ ; i.e.  $C(X) \in \mathcal{C}$ .
- Often we are more interested in estimating the *probabilities* that  $X$  belongs to each category in  $\mathcal{C}$ .

For example, it is more valuable to have an estimate of the probability that an insurance claim is fraudulent, than a classification fraudulent or not.

# Classification: hard vs soft predictions

- **Soft prediction:** estimate a probability
  - $\hat{p}(X) = P(Y = 1 \mid X)$
- **Hard prediction:** turn that probability into a class label
  - $\hat{y}(X) = \mathbf{1}\{\hat{p}(X) > t\}$  for some threshold  $t$

# Why soft predictions matter in credit risk

- We can **rank** accounts by risk (scorecards)
- We can choose different actions at different probability ranges
  - e.g., approve / approve-with-conditions / manual review / decline
- Thresholds change when costs or objectives change

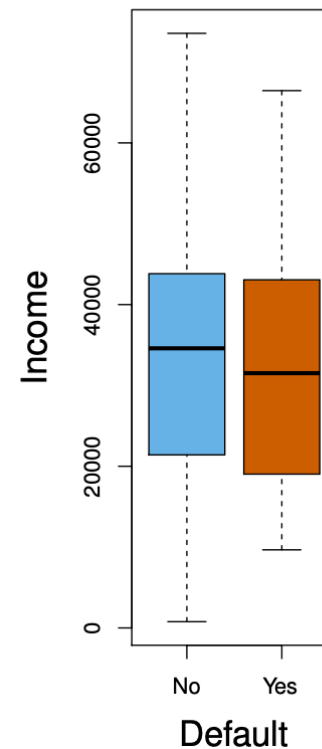
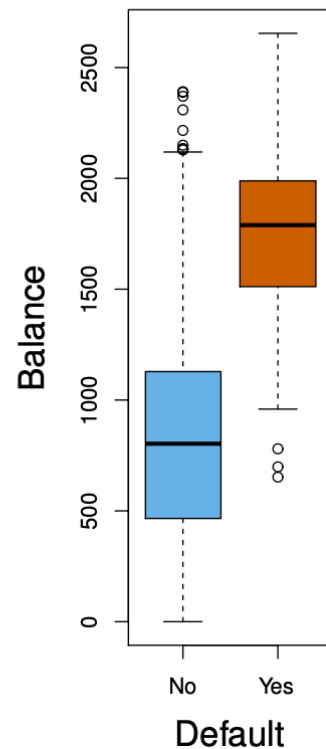
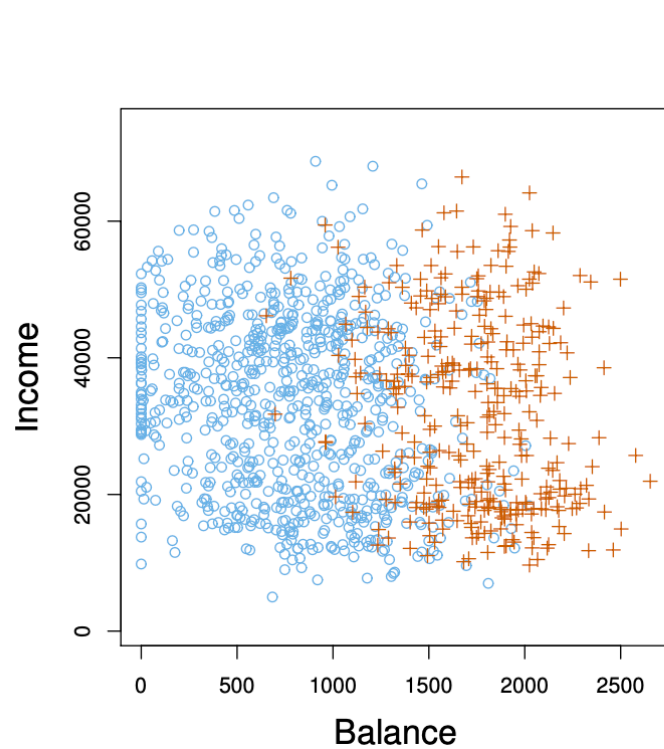
# Thresholds are business decisions

- Picking  $t = 0.50$  is **not** automatically “right”
- You pick  $t$  based on tradeoffs:
  - false positive vs false negative costs
  - capacity constraints (manual reviews)
  - regulatory / policy constraints

# Example: credit card / default-style framing

- $Y = 1$ : "bad outcome" (fraud / default / delinquency / downgrade)
- $X$ : features available at decision time (balance, utilization, income, behavior, etc.)
- Output:  $\hat{p} = P(Y = 1 \mid X)$

## Example: Credit Card Default



## Can we use Linear Regression?

Suppose for the **Default** classification task that we code

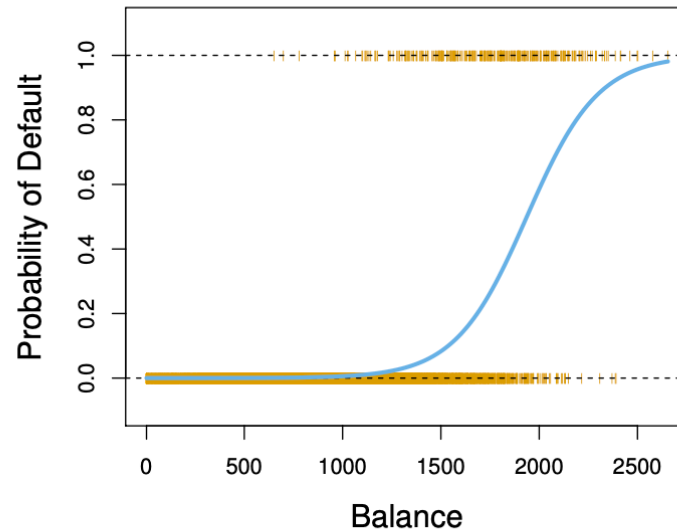
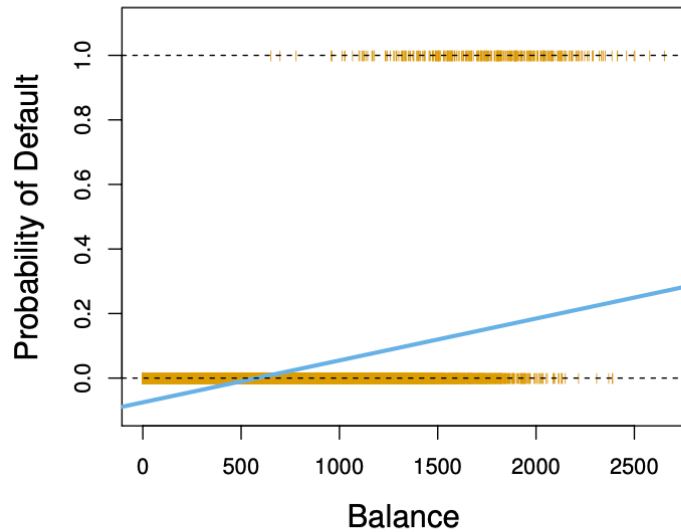
$$Y = \begin{cases} 0 & \text{if No} \\ 1 & \text{if Yes.} \end{cases}$$

Can we simply perform a linear regression of  $Y$  on  $X$  and classify as **Yes** if  $\hat{Y} > 0.5$ ?

- In this case of a binary outcome, linear regression does a good job as a classifier, and is equivalent to *linear discriminant analysis* which we discuss later.
- Since in the population  $E(Y|X = x) = \Pr(Y = 1|X = x)$ , we might think that regression is perfect for this task.
- However, *linear* regression might produce probabilities less than zero or bigger than one. *Logistic regression* is more appropriate.



# Linear versus Logistic Regression



The orange marks indicate the response  $Y$ , either 0 or 1. Linear regression does not estimate  $\Pr(Y = 1|X)$  well. Logistic regression seems well suited to the task.

# Why not just use linear regression?

- Linear regression can predict values  $< 0$  or  $> 1$
- The relationship between probability and predictors is rarely linear
- Classification wants a model constrained to  $[0, 1]$

# Linear vs logistic (big idea)

- Logistic regression models probability using an S-shaped curve
- Think: "linear model on log-odds"

$$\log\left(\frac{p(X)}{1 - p(X)}\right) = \beta_0 + \beta_1 X_1 + \cdots + \beta_p X_p$$

# Fitting logistic regression in R

Small snippet only — we'll do the real work in the [.R](#) demo file.

```
1 fit <- glm(default ~ balance + income + student,  
2           data = train,  
3           family = binomial)  
4  
5 p_hat <- predict(fit, newdata = test, type = "response")
```

# Interpreting coefficients

- Sign: increases vs decreases risk
- Magnitude: relates to odds, not raw probability
- Optional: odds ratio for a 1-unit increase in  $X_j$  is  $\exp(\beta_j)$

# Prediction = probability, not a decision

- Model output:  $\hat{p}_i = P(Y_i = 1 \mid X_i)$
- Decision rule needs a threshold  $t$ :
  - predict 1 if  $\hat{p}_i > t$

# Part 1 wrap

You should be able to:

- explain why logistic regression is used for 0/1 outcomes
- fit a simple logistic regression with `glm()`
- generate predicted probabilities with `predict(..., type="response")`

Next (after break): how to evaluate + choose thresholds.